

COMP6714: Information Retrieval and Web Search

Introduction

Outline

- Introduction of the course
- Introduction of IR and Web Search

Teaching Team

- Lecturer in charge
 - Raymond Wong
 - Email: wong@cse.unsw.edu.au
 - For course related matters, pls send to: cs6714@cse.unsw.edu.au
 - **Took over this subject fr 22T3**
- Previous Lecturers
 - Dr. Yifang Sun
 - Prof. Wei Wang
- Tutors
 - Nicole Lam, Shaowen Tang, Dominic Wong (Volunteer)

Lectures

- Lecture slides
 - From webcms: <http://www.cse.unsw.edu.au/~cs6714>
- Live lectures
 - Tue 14:05-15:55
 - Topic overview, examples, Q&A
 - Likely 1-2hr depending on Q&A
- Recorded topic videos
 - Released each week
- Where?
 - Links to the live lecture & the videos can be found **on Moodle**

Forum discussions & Online consultations

- We use **Ed** for forum Q&A
 - If you haven't had access to Ed, e.g., late enrolment, please email: cs6714@cse
- Online consultations
 - via Blackboard Collaborate on Thu, Fri (TBC)
 - Check the timetable on webcms for details
 - Any questions related to the course content or cs6714 matters

Communications...in summary

- Q&A on lectures/assignments/etc - **Ed forum**
 - **Participation** in answering some of the questions will be considered when we scale your final course score at the end
- Help/long-questions on lectures/assignments - online **consultations**
- Individual feedback / admin issues (cannot access Ed) - **email cs6714@cse**
- Any assignment special considerations - check & contact **“UNSW Special Consideration”**

How this course will be run this year

- Second time I run this course
- Stick with the same lecture schedule and slides
- Overall, it will be similar to the previous offerings except:
- More applied and less theoretical:
 - Less “proofing” in the assignment/exam
 - Increase the programming project weight
- Bring the assignment forward (earlier)

Course Aims

- Not possible to cover every aspect of Information Retrieval and Web Search
- We will focus on
 - concepts
 - algorithms
 - principles
- We will not focus on
 - programming languages and API
 - specific platforms/tools
- Make use of tutorials and documents on the Internet

Expectation

- What are expected in this course
 - Many modules covering a **broad spectrum** of IR/NLP/SE
 - Heavy workload expected: must read and digest the **textbook and slides + additional notes**
 - Requires substantial **algorithm/data structure** design/analysis experience & capability + some maths.
 - **Up-to-date** viewpoints, understanding, knowledge (from the academia & industry)
 - ➔ Plan your time well
- You are welcome to ask questions at anytime
- Review after the lecture

Knowledge Assumed (non-exhaustive)

- Data structures & algorithms:
 - Heap/priority queue: build a heap in $O(n)$ time?
 - Membership query: tradeoffs? worst/avg-case time complexities = ?
 - Recursion:
 - DFS/BFS/Best-first search

Given an array A of integers. Design an algorithm to return two elements x, y in A , such that $x + y = 100$ if any, and

1. the algorithm takes $O(n \cdot \log(n))$ time, or
2. the algorithm takes $O(n)$ time

Knowledge Assumed (non-exhaustive)/2

- C/C++ & Python Programming:
 - Pointer
 - `sizeof(int) = ?` `sizeof(p) = ?` `sizeof(*p) = ?`
`sizeof(str) = ?`
 - Be able to learn to use new Python libraries and write & debug python programs
 - Quickly learn a python-based framework in this course

Knowledge Assumed (non-exhaustive)/3

- CS Architecture
 - Memory hierarchy: name the levels?
 - Bit representation: binary string for any x ? How to obtain the 3rd-5th bits of a byte?
- Maths
 - Calculus: How to find the minimum/minimal value of a function $f(x)$?
 - Probabilities and statistics: rv; linearity of expectation; indicator variable; number of heads by tossing a biased coin n times; Bayesian theorem
 - Linear algebra: inner/dot product of $\mathbf{u}$ and $\mathbf{v} = ?$
matrix multiplication

Assessment

```
asgt      = mark for the assignment      (out of 10)
proj      = mark for the project          (out of 40)
exam      = mark for final exam          (out of 50)
okEach    = exam > 20                    (after scaling)
mark      = asgt + proj + exam
grade     = HD|DN|CR|PS  if mark >= 50 && okEach
           = FL          if mark < 50 && okEach
           = UF          if !okEach
```

Assignment & Project

- Non programming assignment
- Programming project
- Late submission: 5% of the max subtracted from earned marks per day (no acceptance after 5 days late) – *see Course Outline for details.*

Final exam

- One final exam (worth 50 %)
- If you are ill on the day of the exam, do not attend the exam – c.f. UNSW fit-to-sit policy. Apply for special consideration asap.
- It'll be an in-person exam. More details to be provided later in the course.

Python examples/exercises (using the github link from webcms)

- From previous offerings
- Warm-up preparation for the project
 - L0 – python3 & jupyter setup
 - During week 1
 - L1 – Inverted index – list intersection
 - After week 1
 - L2 – Preprocessing (e.g., tokenization)
 - After week 2
- Other exercises may be relevant, and will be recommended in specific lectures
 - You will also be busy on your assignment & project after week 3

Warning

- This course has
 - Broad coverage
 - Heavy workload
 - High fail rate $\geq 20\%$ *aim: $< 10\%$*
- Specially, we do not accept personal plea or excuses
 - if you have valid reasons that affect your performance, apply for a UNSW Special Consideration
 - <https://student.unsw.edu.au/special-consideration>.

Warning - cont.

- Common excuses/arguments

- I spent so much time and effort on this course but still failed?
- I did the work by myself and may have shared it with my classmate for discussion.
- If I fail this course, I will [...]. Please.

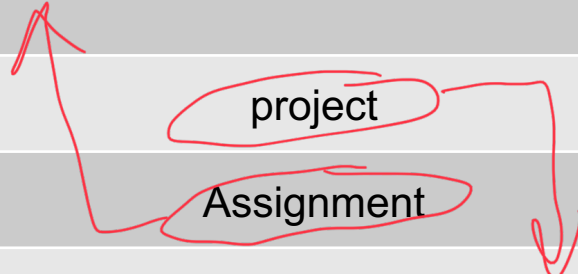
Academic honesty and plagiarism

- Zero tolerance to plagiarism
 - You will get 0 marks
- Examples of misconduct:
 - Copy other students' work
 - **Let other students copy your work**
 - Copy from GitHub
 - Find a ghost writer
 - ...
- I will not accept the following excuses:
 - “I’ve left the lab with my screen unlocked”
 - “He stole it from my computer”
 - “I only gave my code to A. A didn’t use it but gave it to B”
 - ...
- Make sure you read all types of plagiarism, esp. collusion in <https://student.unsw.edu.au/plagiarism>.

Please do not enrol if you...

- Don't have the required knowledge
 - Cannot produce correct Python program on your own
 - Have poor time management
 - Are too busy to watch lecture videos
 - Are not honest
-
- Otherwise, you are likely to perform badly in this subject

Tentative course schedule

Week	Topic	Labs/Assignment/Project
1	Course Introduction + Boolean Retrieval	
2	Preprocessing	
3	<u>Index construction</u>	
4	<u>Compression</u>	
5	Vector Space Model	
6	Flexibility Week (no lecture)	 project Assignment
7	Evaluation	
8	Crawling	
9	Link Analysis	
10	Revision and Exam Preparation	

COMP6714 vs COMP9319

- COMP6714 – Information Retrieval
 - Search basics, search quality
 - Python
- COMP9319 – Web Data Compression and Search
 - Size, performance & scalability
 - C/C++

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Cut down
in 6714

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What is Information Retrieval?

- Let's start with **Search**
- Search on the Web is a daily activity for many people throughout the world
- Search and communication are most popular uses of the computer
- Applications involving search are everywhere
- The field of computer science that is most involved with R&D for search is *information retrieval (IR)*

Information Retrieval

- Information Retrieval (IR) is **finding material** (usually documents) of an **unstructured** nature (usually text) that satisfies an **information need** from within **large collections** (usually stored on computers).
- These days we frequently think first of **web search**, but there are many other cases:
 - E-mail search
 - Searching your laptop
 - Corporate knowledge bases
 - Legal information retrieval
- Primary focus of IR since the 50s has been on *text* and *documents*

Search Engine

- As a user
 - What you can do?
 - What to expect?
- As a server
 - How to meet the users' requirements?
 - How to improve the users' experience?
- As an observer
 - How to evaluate a search engine?

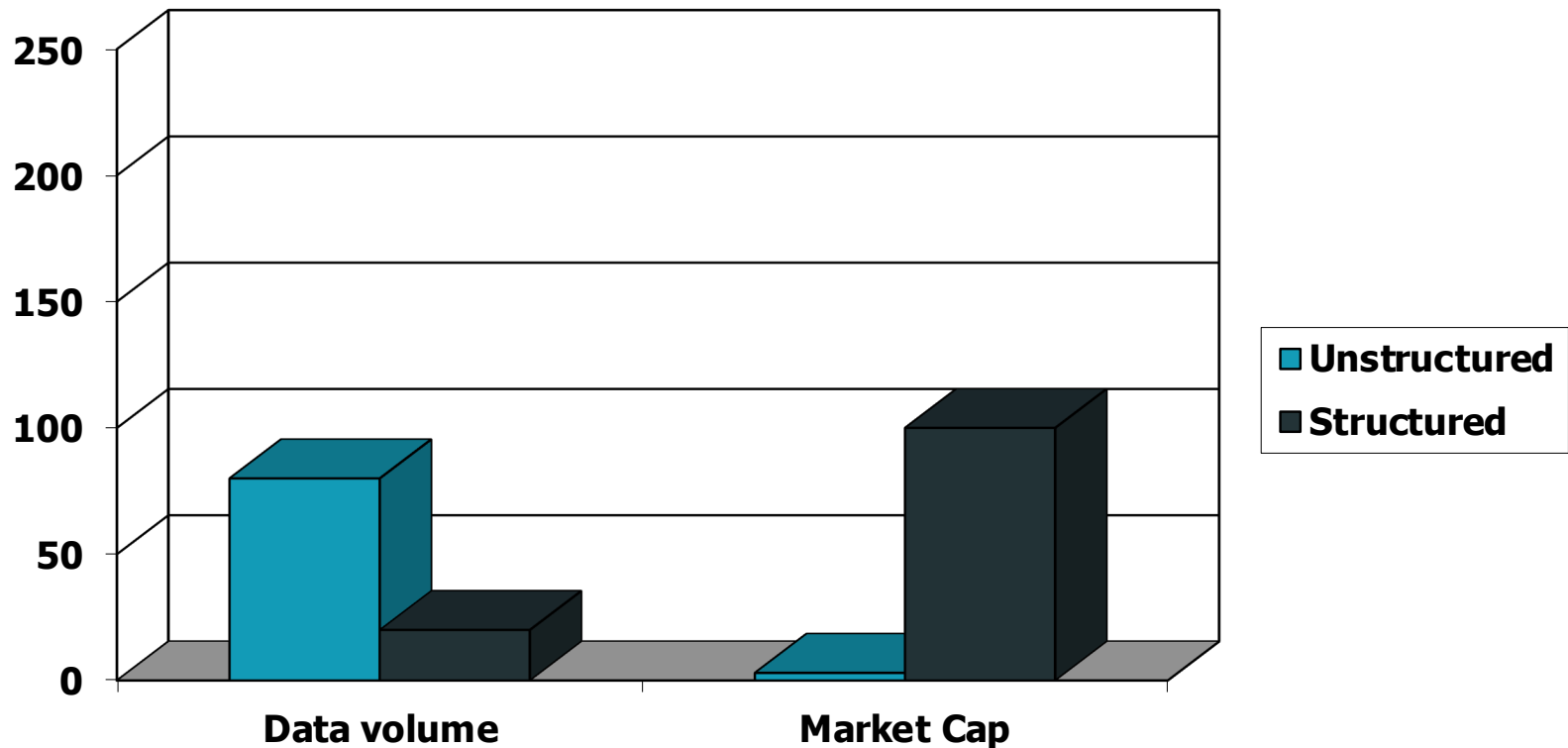
Basic assumptions of Information Retrieval

- **Collection**: A set of documents
 - Assume it is a static collection for the moment
- **Goal**: Retrieve documents with information that is **relevant** to the user's **information need** and helps the user complete a **task**

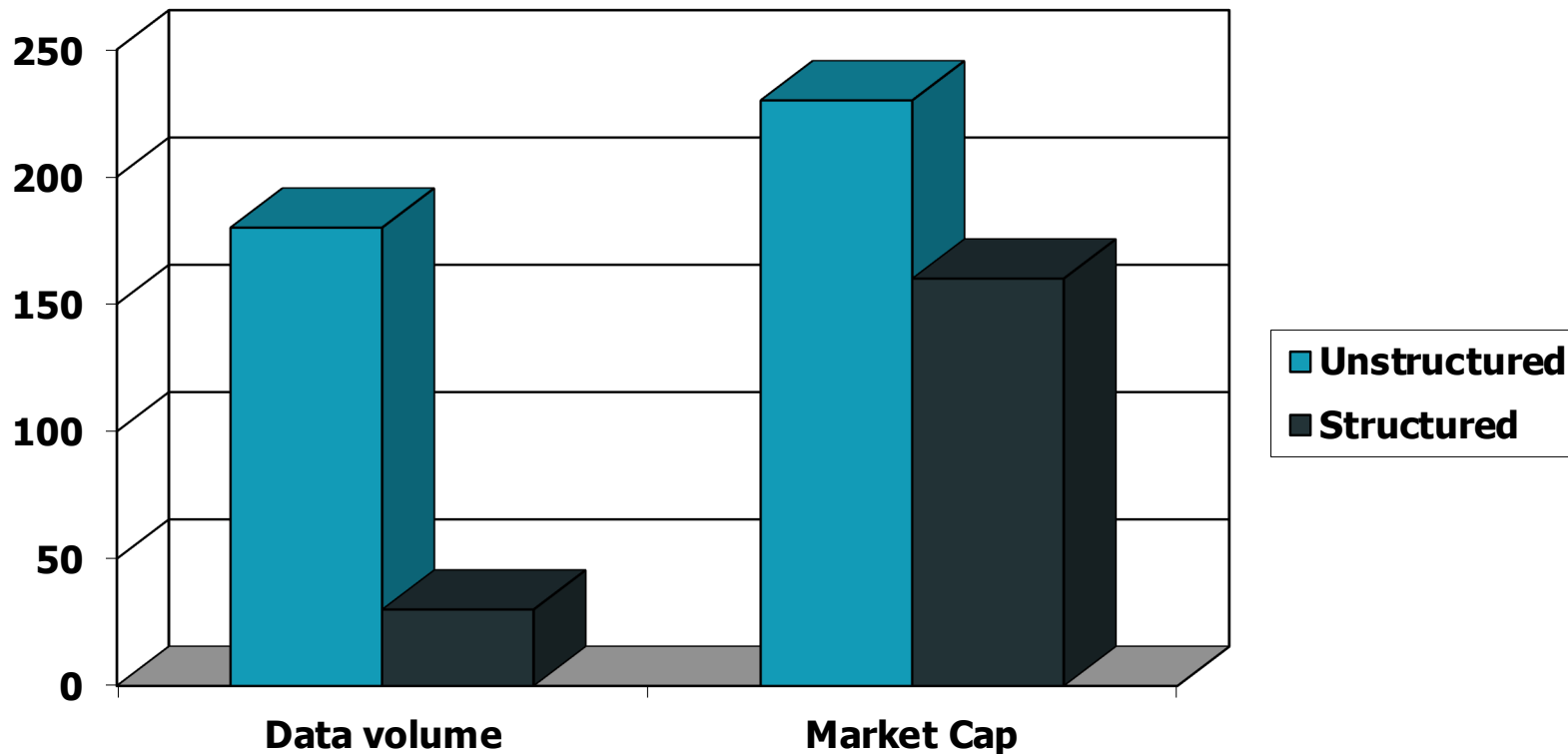
What is a Document?

- Examples:
 - web pages, email, books, news stories, scholarly papers, text messages, Word™, Powerpoint™, PDF, forum postings, patents, IM sessions, etc.
- Common properties
 - Significant text content
 - Some structure (e.g., title, author, date for papers; subject, sender, destination for email)

Unstructured (text) vs. structured (database) data in the mid-nineties



Unstructured (text) vs. structured (database) data in 2019



Documents vs. Database Records

- Database records (or *tuples* in relational databases) are typically made up of well-defined fields (or *attributes*)
 - e.g., bank records with account numbers, balances, names, addresses, social security numbers, dates of birth, etc.
- Easy to compare fields with well-defined semantics to queries in order to find matches
- Text is more difficult

Documents vs. Records

- Example bank database query
 - *Find records with balance > \$50,000 in branches located in Amherst, MA.*
 - Matches easily found by comparison with field values of records
- Example search engine query
 - *bank scandals in western mass*
 - This text must be compared to the text of entire news stories

Comparing Text

- Comparing the query text to the document text and determining what is a good match is the core issue of information retrieval
- Exact matching of words is not enough
 - Many different ways to write the same thing in a “natural language” like English
 - e.g., does a news story containing the text “*bank director in Amherst steals funds*” match the query?
 - Some stories will be better matches than others

Dimensions of IR

- IR is more than just text, and more than just web search
 - although these are central
- People doing IR work with different media, different types of search applications, and different tasks

Other Media

- New applications increasingly involve new media
 - e.g., video, photos, music, speech
- Like text, content is difficult to describe and compare
 - text may be used to represent them (e.g. tags)
- IR approaches to search and evaluation are appropriate

Dimensions of IR

Content	Applications	Tasks
Text	Web search	Ad hoc search
Images	Vertical search	Filtering
Video	Enterprise search	Classification
Scanned docs	Desktop search	Question answering
Audio	Forum search	
Music	P2P search	
	Literature search	

IR Tasks

- Ad-hoc search
 - Find relevant documents for an arbitrary text query
- Filtering (aka information dissemination)
 - Identify relevant user profiles for a new document
- Classification
 - Identify relevant labels for documents
- Question answering
 - Give a specific answer to a question

Big Issues in IR

- Relevance

- What is it?
- Simple (and simplistic) definition: A relevant document contains the information that a person was looking for when they submitted a query to the search engine
- Many factors influence a person's decision about what is relevant: e.g., task, context, novelty, style
- *Topical relevance* (same topic) vs. *user relevance* (everything else)

Big Issues in IR

- **Relevance**
 - **Retrieval models** define a view of relevance
 - **Ranking algorithms** used in search engines are based on retrieval models
 - Most models describe statistical properties of text rather than linguistic
 - i.e. counting simple text features such as words instead of parsing and analyzing the sentences
 - Statistical approach to text processing started with Luhn in the 50s
 - Linguistic features can be part of a statistical model

Big Issues in IR

- Evaluation
 - Experimental procedures and measures for comparing system output with user expectations
 - Originated in Cranfield experiments in the 60s
 - IR evaluation methods now used in many fields
 - Typically use *test collection* of documents, queries, and relevance judgments
 - Most commonly used are TREC collections
 - *Recall* and *precision* are two examples of effectiveness measures

Web Search

- New Challenges
- How to obtain data?
- Additional features for Web data

Unranked retrieval evaluation:

Precision and Recall

- **Precision:** fraction of retrieved docs that are relevant
 $= P(\text{relevant} | \text{retrieved})$
- **Recall:** fraction of relevant docs that are retrieved
 $= P(\text{retrieved} | \text{relevant})$

	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision $P = tp / (tp + fp)$
- Recall $R = tp / (tp + fn)$

